

◁ **HIDDEN CAVE**

This underground lake in Chapada Diamantina National Park, Brazil, contains some of the continent's oldest rocks.

◁ **RUPTURING VOLCANO**

The 16,480-ft- (5,023-m-) high Tungurahua volcano in Ecuador is one of many towering volcanoes that have burst through the Andes.

◁ **CONTINENTAL SPINE**

The Andes are the world's longest mountain range, stretching 4,500 miles (7,200 km) - almost along the entire western edge of South America.

◁ **CHANGED COURSE**

The Amazon River is the world's largest river, carrying huge amounts of water through its enormous, lush lowland basin. Until 10 million years ago, it used to be a huge inland lake.

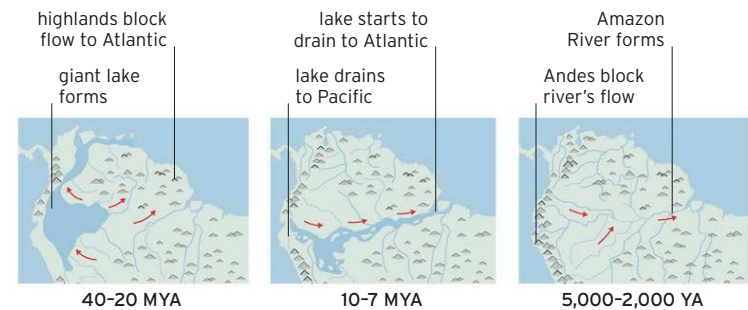
Nazca Plate was driven down into the mantle, a deep trench opened up in the ocean floor along the west coast of South America. The descending Nazca Plate dragged against the South American Plate, fracturing and deforming it.

The Andes started to form 50 million years ago as rocks were crumpled and piled on top of each other. Meanwhile, the Nazca Plate, melting as it was pushed into the mantle, caused magma to punch through the young mountain range as volcanoes, adding a series of towering volcanic peaks such as Cotopaxi in Venezuela, and Chimborazo and Tungurahua in Ecuador.

River basins

Between the three ancient shields and the newer fold mountains of the Andes are lowland river basins, filled over time with layer upon layer of sediment. There are three main basins, forming a linked lowland belt running north to south down the middle of the continent. The Orinoco in the north flows from the Venezuelan Andes into the Atlantic south of Trinidad. The sprawling Amazon River in the center flows right across South America from the Andes to emerge in the Atlantic at the equator. Until about 10 million years

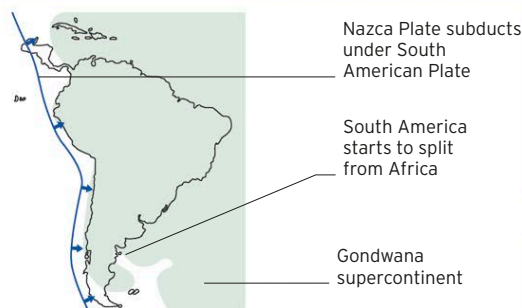
ago, the Amazon pooled into a giant, inland lake and drained westward into the Pacific Ocean. The Chaco-Pampa in the south is a large basin through which the Parana, Paraguay, and other rivers flow into the Rio de la Plata.

**HOW THE AMAZON BASIN FORMED**

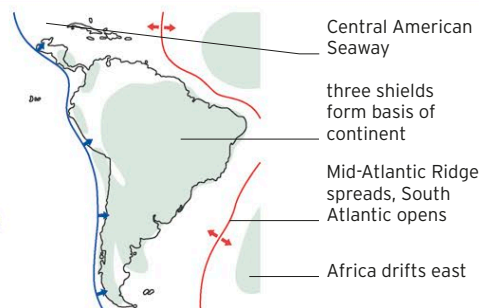
The Amazon River flowed from the Guiana Highlands into a huge lake, which drained into the Gulf of Mexico and the Pacific Ocean. Eventually, the rising Andes forced the lake to drain into the Atlantic Ocean instead.

BUILDING THE CENTRAL AND SOUTH AMERICAN CONTINENT

KEY — Convergent boundary — Divergent boundary



94 MILLION YEARS AGO South America begins to break away from Africa and move west. The South American Plate abuts the Nazca Plate, which subducts beneath it.



50-40 MILLION YEARS AGO The Mid-Atlantic Ridge spreads, forcing South America and Africa farther apart and creating the Atlantic Ocean. The Central American Seaway separates the Americas.



18,000 YEARS AGO The Central American Seaway, which once separated the continent from Central America, has now closed and South and North America have become linked by land.